Seeing is Believing? How Learning Modes Shape Belief Bias and Discrimination

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Outline for Today

1. Motivation

Motivation: The Puzzle of Persistent Bias

- *Biased Beliefs*: an important driver of discrimination (Bohren, et al., 2023; Mengel and Campos-Mercade, 2024)
- Information is not very effective at reducing bias (e.g. Wozniak et al., 2019; Pedersen & Nielsen, 2024)
- Why do biased beliefs persist despite abundant information?

Motivation: A Missing Piece - How People Gather Information

- The Usual Suspect: biased belief updating (e.g. Mobius, et al., 2014; Coutts, 2019)
- Belief Formation is a 2-part process
  1. Acquire Information
  2. Interpreting Information
- Missing Piece: **Mode of Information Acquisition**

Question: How does the *mode of information acquisition* affect beliefs and subsequent discrimination?

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Modes of Information Acquisition: A Concrete Example

Two employers want to learn about productivity of candidate groups

- Employer A: Rolling review, stop reviewing resumes when satisfied
  → **Endogenous Information Acquisition**
- Employer B: Fixed review window, review all collected resumes
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Research Question

- How does the way we acquire information shape belief bias and subsequent discrimination?
- I study 2 modes of acquisition
  - Endogenous: Stop-at-will
  - Exogenous: Fixed sample size
- Lab experiment where employers evaluate & hire workers from 2 groups

Preview of Findings

- Endogenous acquisition sustains biased beliefs and discrimination
- Exogenous acquisition eliminates bias with similar amount of information
- Mechanism: realization-based stopping produces skewed evidence

Contributions

- Bridge gap in literature: how mode of information acquisition shapes beliefs
- Identify novel source of persistent bias: endogeneity of information acquisition
- Shed light on key mechanism: realization-based stopping
- Propose & validate practical interventions

2. Experimental Design

Experiment Structure

- Setup: Employers evaluate and hire workers
- Workers: Asian Group & Hispanic Group (100 Workers each)
  - Domain: 12-question math test
  - Productivity: Math score
  - Objective Benchmark: **Equal productivity distribution**

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- Employers: Separate hiring process into 2 stages:
  1. Learning: Information acquisition & belief updating
  2. Hiring: Wage offers
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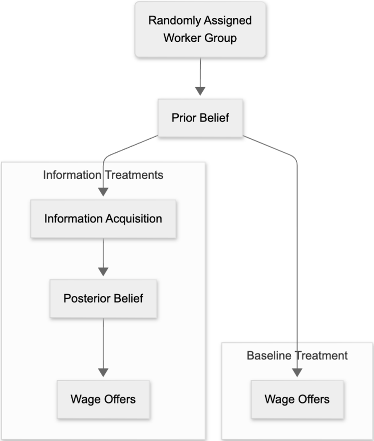
Experiment Structure: Flowchart



Go Back

Start of Experiment: Worker Group Assignment

1. Each employer randomly assigned to **one** worker group (*between-subject*)
2. Learns about worker group, math test, and scoring

Your Role

- You are an **employer** hiring from a group of **100 Asian workers** from Prolific.
- They will be referred to as the worker group throughout the survey.

About the Workers

- These workers completed an **SAT-level math test** with **12 questions** in **6 minutes**.
  - Each worker's **score** = number of correct answers
  - Click here to see sample questions

- Report belief on score distribution in the group
- Incentivized for accuracy
- Identical for all treatments

Out of the 100 Hispanic workers, how many do you think have a score of... (the numbers should add up to 100)

1?	0
2?	0
3?	0
4?	0
5?	0
6?	0
7?	0
8?	0
9?	0
10?	0
11?	0
12?	0
Total	0

Common Elements:

- Sequential random sampling without replacement
- Button click $\rightarrow$ random worker $\rightarrow$ observe score bin
- 3 second cool-down

Sampling Interface

Press the button below to draw workers.

Draw a Worker

Worker 1 score: between 7 to 9
Worker 2 score: between 7 to 9
Worker 3 score: between 10 to 12

Treatment Variation:

1. **Voluntary**: can stop/continue any time
2. **Exogenous Mean**: draw fixed number
    - Equals average sample size from Voluntary
    - 10 and 8 for Hispanic and Asian
3. **Exogenous 20**: make 20 draws

Voluntary Instructions

1. In Part 2, you can learn about worker performance by randomly drawing workers. You will then make the evaluations again.
    1. Click the **"Draw a Worker"** button to reveal a randomly selected worker's **score range** (1–3, 4–6, 7–9, or 10–12).
    2. Each worker is equally likely to be selected.
    3. You can draw a worker **every 3 seconds**.
    4. You can draw as many workers as you want

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Voluntary Sampling Example

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Draw a Worker

Worker 1 score: between 10 to 12
Worker 2 score: between 4 to 6
Worker 3 score: between 7 to 9
Worker 4 score: between 7 to 9
Worker 5 score: between 1 to 3
Worker 6 score: between 1 to 3

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   4. Continue until you have drawn **exactly 8** workers.

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Exogenous Mean Sampling Example

Press the button below to draw workers.

Draw a Worker

Worker 1 score: between 4 to 6
Worker 2 score: between 7 to 9
Worker 3 score: between 7 to 9
Worker 4 score: between 10 to 12
Worker 5 score: between 4 to 6
Worker 6 score: between 4 to 6
Worker 7 score: between 4 to 6
Worker 8 score: between 4 to 6
You have finished drawing workers!

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   2. Each worker is equally likely to be selected.
   3. You can draw a worker **every 3 seconds**.
   4. Continue until you have drawn **exactly 20** workers.

- Same interface as prior elicitation (incentivized)

Out of the 100 Hispanic workers, how many do you think have a score of... (the numbers should add up to 100)

1?	0
2?	0
3?	0
4?	0
5?	0
6?	0
7?	0
8?	0
9?	0
10?	0
11?	0
12?	0
Total	0

- Employers see a sequence of 10 randomly selected workers
- Each worker has a "resume"
- Worker ethnicity conveyed by avatar and nickname

Example Hispanic worker profile:

Nickname:	Marcos
Gender:	Male
Country:	United States
Age:	18-45
Favorite color:	purple

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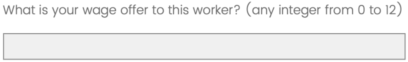
- Worker ethnicity conveyed by avatar and nickname

Example Asian worker profile:

Nickname:	Tian
Gender:	Male
Country:	United States
Age:	18-45
Favorite color:	purple

- Makes an incentivized wage offer to each worker
- Dominant Strategy: Wage offer = true valuation

Wage offer interface:



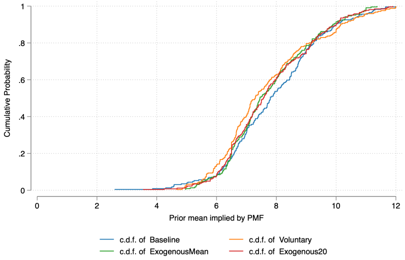
Implementation

- Pre-registered on AsPredicted, IRB approval from UC Davis
- Survey programmed in Qualtrics, conducted on Prolific
- Around 1600 employers
- Average time: 15 minutes
- Average earnings: \$3.72, hourly rate = \$15
- 2/3 of employers answered comprehension quiz correctly on first try

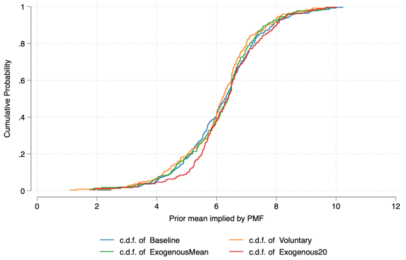
3. Results

Distribution of Prior Mean Beliefs

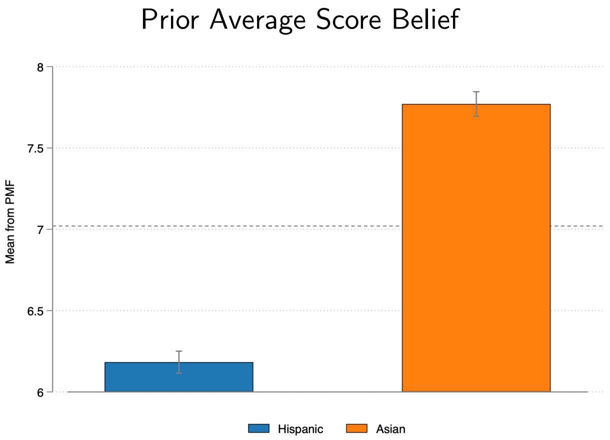
Asian Workers



Hispanic Workers



Prior Beliefs Biased Against Hispanics



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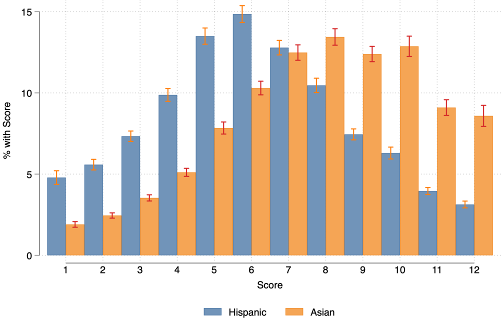
Prior Belief Distribution



Elicitation Interface

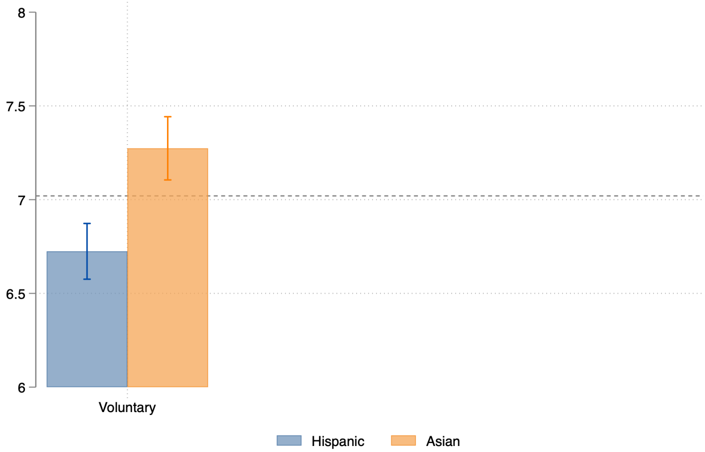
Out of the 100 Hispanic workers, how many do you think have a score of... (the numbers should add up to 100)

1?	0
2?	0
3?	0
4?	0
5?	0
6?	0
7?	0
8?	0
9?	0
10?	0
11?	0
12?	0
Total	0

Priors Leave Scope for Learning

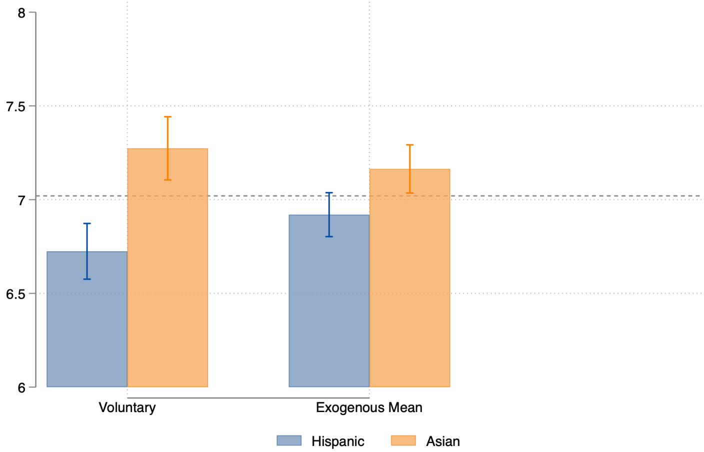
- Priors biased against Hispanics & far from truth
- Next: learn about group productivity
- Incentives: learn accurately
- How well do they learn under each information acquisition mode?

Endogenous Information Acquisition Sustains Bias



Voluntary Draws
By Sample Size

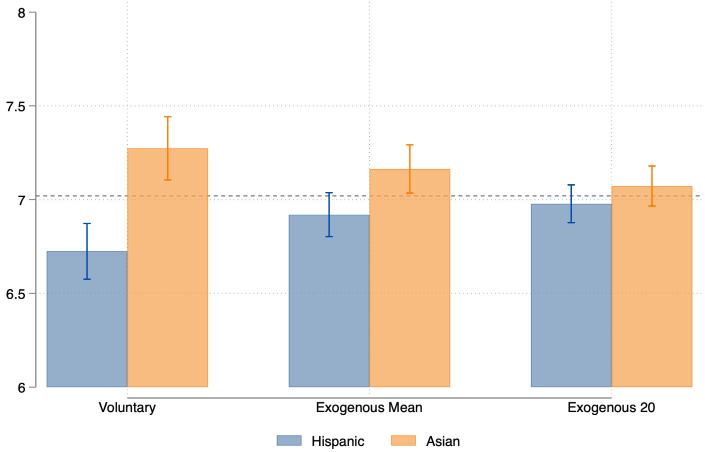
Exogenous Information Acquisition Reduces Bias



Voluntary Draws

By Sample Size

Exogenous Acquisition with 20 Draws Eliminates Bias



Voluntary Draws
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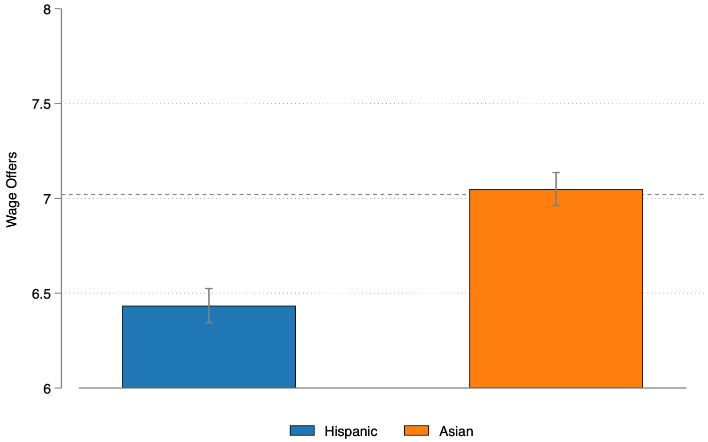
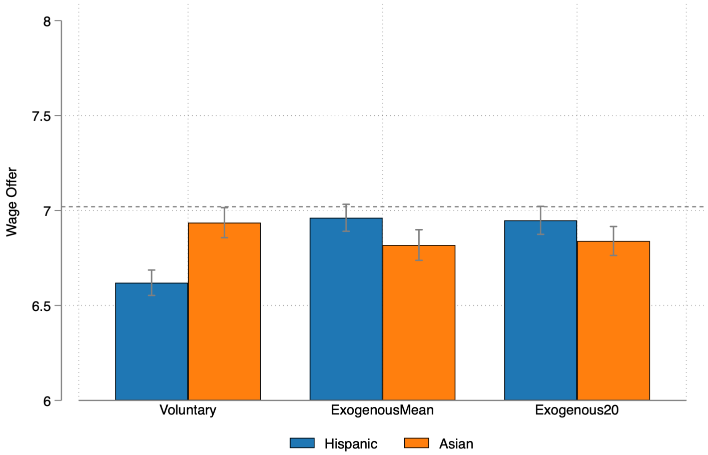
Baseline: Wage Discrimination Against Hispanics

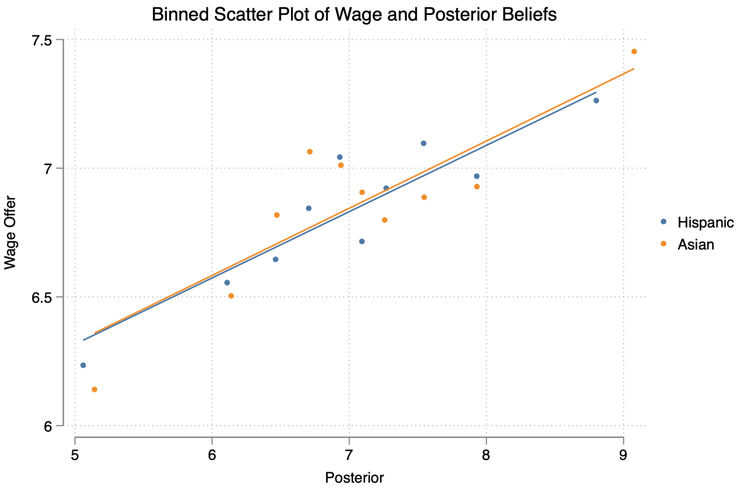


Flowchart

Wage Discrimination After Information Acquisition



Discrimination Driven by Biased Beliefs



Taking Stock

- Endogenous information acquisition sustains biased beliefs
- Exogenous acquisition reduces bias with similar amounts of information
- Wage offers closely track beliefs
- Discrimination is mostly driven by biased beliefs

4. Mechanisms

Mechanisms: 2 Possibilities

Question: Why does endogenous information acquisition lead to persistent bias?

1. Bias-Sample Size Correlation ▸ Theory

2. Realization-Based Stopping
   - E.g. Low Score → High Score → High Score → Stop (Asian)
   - Skews evidence

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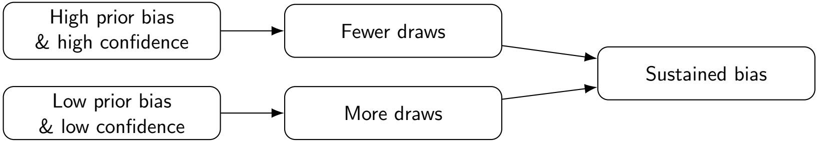

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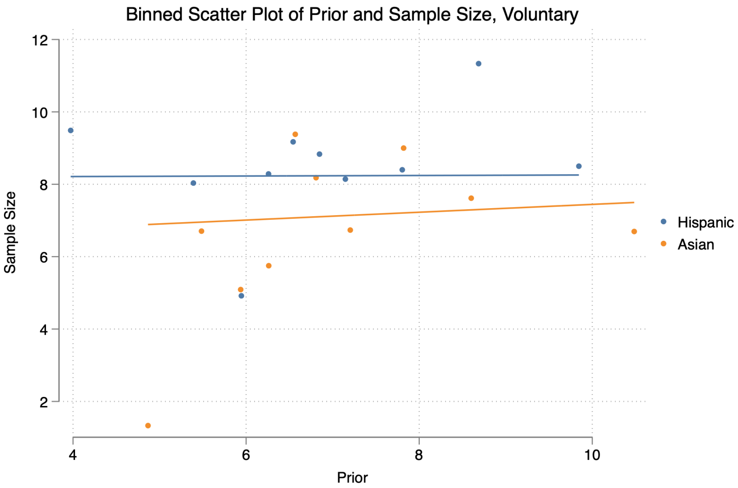
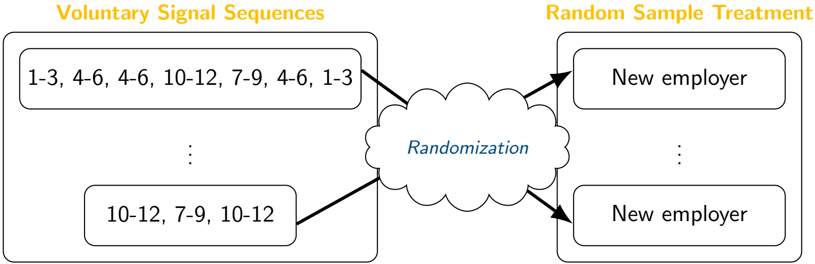
What is the Correlation Between Prior and Sample Size?



Random Sample Treatment: Testing the Bias-Sample Size Link

- How to test the role of Bias-Sample Size Correlation? Break the link.



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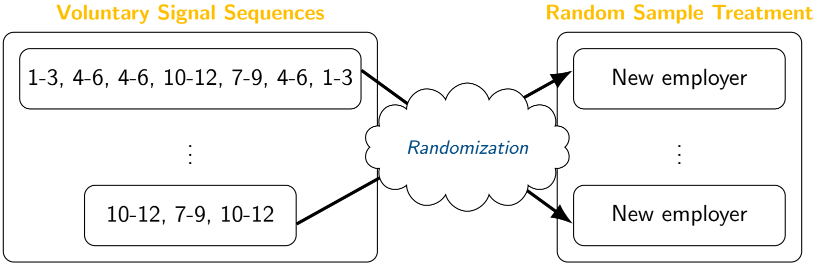

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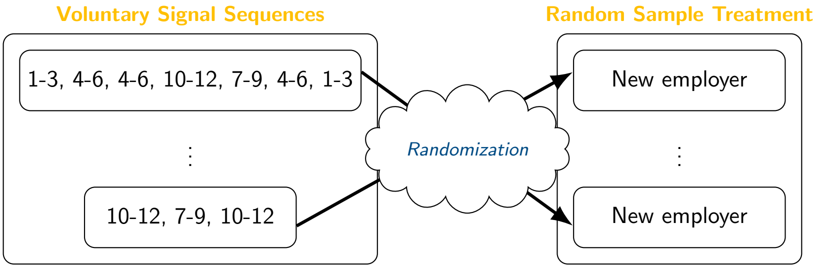

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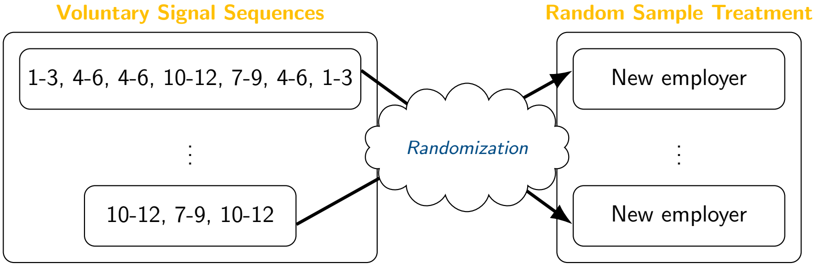

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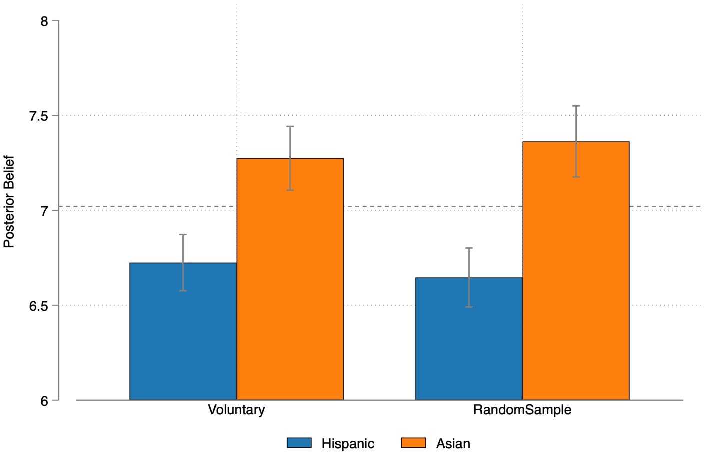
Bias-Sample Size Correlation: Not A Major Mechanism



Testing Realization-Based Stopping: Commitment Treatment

- What is the role of realization-based stopping?
- **Commitment** Treatment: remove realization-based stopping

The Commitment Treatment

- Employers commit ex-ante to the number of workers they will draw
- They *must* draw that number
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Here you will draw 5 workers from the group. The "Next" button will appear after you finish drawing workers. You will be able to refer to the drawn workers again when you make your score estimates.

Press the button below to draw workers.

Draw a Worker

Worker 1 score: between 10 to 12
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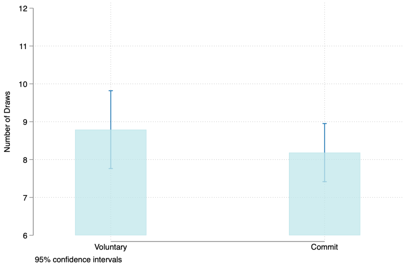
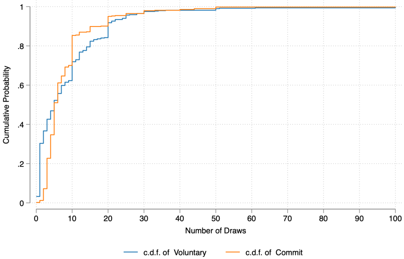
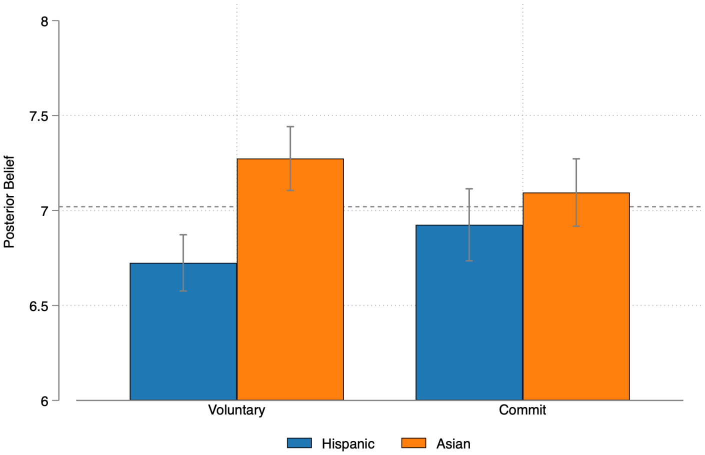
Number of Draws in Voluntary vs. Commitment





Realization-Based Stopping Leads to Belief Distortions



Estimating the Empirical Stopping Rule

- So far: *Realization-Based Stopping* is key mechanism
- Taking to the data: Estimate empirical stopping rule in **Voluntary**
- Logit model on sampling data, separately for Asian & Hispanic

$$\Pr(\text{Stop}_{it} = 1) = \Lambda\big(\alpha + \beta_1\,\text{DistPrior}_{it} + \beta_2\,\text{HighScore}_{it} + \beta_3\,\text{Prior}_i + \beta_4\,\text{t10}_t + \beta_5\,\text{t20}_t\big)$$

where $\Lambda(z) = \frac{e^z}{1+e^z}$ is the logistic CDF.

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Realization-Based Stopping in Action

	Asian	Hispanic
$\lvert$Sample Mean − Prior$\rvert$	0.142***	0.060
	(0.046)	(0.041)
High Score	0.279***	0.057
	(0.107)	(0.105)
Prior Belief	0.081*	-0.041
	(0.041)	(0.041)
10th Draw Dummy	1.399***	1.026***
	(0.180)	(0.183)
20th Draw Dummy	2.050***	1.882***
	(0.237)	(0.239)
Number of Subjects	233	234
Clustered SE	Yes	Yes

5. Conclusion

Key Takeaways

- Endogenous information acquisition sustains biased beliefs
- Biased beliefs in turn leads to wage discrimination
- Key mechanism: realization-based stopping leads to biased information
- **Voluntary** employers leave money on the table

Policy Implications

- The way people acquire information matters
- Structuring information acquisition to reduce endogeneity can be effective at reducing bias in evaluations
- Provides empirical support for structured review windows in hiring
- Other applications: performance evaluations, media/news consumption, medical treatment evaluation, etc.

Thank You

Your feedback is appreciated!

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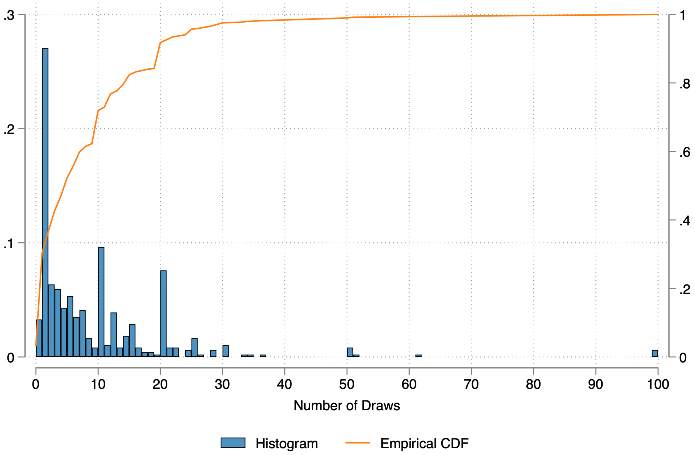
6. Appendix

Sample Sizes in Experience-Voluntary



Voluntary Posteriors

The Conceptual Problem

- An employer is interested in learning about productivity in worker group
- Learning: sequentially drawing scores (i.i.d signals from distribution)
- Goal: make wage offer as close to true productivity (no other info, use group average)

Setup and Parameters

- True group average productivity $\theta \in \mathbb{R}$
- Prior belief distribution: $p_0(\theta) = \mathcal{N}(\mu_0, \sigma_0^2)$
- Discrete time $t \in \{1, 2, \ldots\}$
- In each period, the employer can draw a random score $x_t \sim \mathcal{N}(\theta, \sigma_d^2)$
- Each draw incurs fixed cost $c \geq 0$

The Rational Employer Faces an Optimal Stopping Problem

- In each period, the employer Bayesian updates based on signals
- Decides whether stop or continue
- If continue, repeat
- If stop, earn instrumental utility u:

$$u(\mu_\tau, \theta) = \begin{cases} 1 & \text{if } |\mu_\tau - \theta| \leq b \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Belief Updating | Rational Employer's Problem | Cognitive Consistency

The Rational Employer: Key Insights

1. Posterior variance determines stopping point
2. Threshold rule: stop if posterior variance crosses a threshold
3. Equivalence of Learning Modes: Voluntary = Ex-Ante Commitment

▸ Commitment

Comparative Statics

Optimal stopping time of the rational employer...

- increases in prior variance
- increases with signal noise
- decreases with signal cost

Back to Mechanisms

Belief Updating

- After observing t signals, belief updating using Bayes' Rule
- Posterior belief distribution: $p_t(\theta|D_t) \propto \mathcal{N}(\mu_t, \sigma_t^2)$, where $\sigma_t^2 = \frac{1}{\frac{t}{\sigma_d^2} + \frac{1}{\sigma_0^2}}$

  and $\mu_t = \sigma_t^2 \left(\frac{t\bar{x}}{\sigma_d^2} + \frac{\mu_0}{\sigma_0^2} \right)$.
- In particular,

$$\mu_t = \omega_t \bar{x} + (1 - \omega_t)\mu_0 \tag{2}$$

  where $\omega_t = \frac{t\sigma_t^2}{\sigma_d^2}$

Back to Theory

The Rational Agent's Problem

In period t, the agent's decision problem is captured by the following Bellman Equation:

$$V(\mu_t) = \max \left\{ \underbrace{u(\mu_t, \theta)}_{\text{Stop}}, \underbrace{\mathbb{E}\left[V(\mu_{t+1}) \,\middle|\, \mu_t \right] - c}_{\text{Continue}} \right\} \tag{3}$$

It can be shown that

$$u(\mu_t, \theta) = 2\Phi\left(\frac{b}{\sigma_t}\right) - 1$$

Insight: there is a threshold value of posterior variance $\bar{\sigma}$ such that the agent stops if $\sigma_t^2 \leq \bar{\sigma}$.

Back to Theory

Equivalence of Learning Modes for Rational Agent

Insight: the Sequential Problem and the Ex-Ante Commitment Problem are Equivalent

- Preferences are time consistent
- Utility only depends on posterior variance, which evolves deterministically
- Observed signals does not shift the stopping time

Cognitive Consistency

Incorporating belief utility:

$$U = \underbrace{u(\mu_t, \theta)}_{\text{Instrumental Utility}} + \underbrace{\gamma v(\mu_0, \mu_t)}_{\text{Belief Utility}} \tag{4}$$

where belief utility takes the form

$$v(\mu_0, \mu_t) = -(\mu_0 - \mu_t)^2. \tag{5}$$

The Decision Problem Under Cognitive Consistency

$$\tilde{V}(\mu_t) = \max \left\{ \underbrace{u(\mu_t, \theta)}_{\text{Stop}}, \underbrace{\mathbb{E}[\tilde{V}(\mu_{t+1}) + \gamma v(\mu_0, \mu_{t+1}) | \mu_t] - c}_{\text{Continue}} \right\} \tag{6}$$

- Cognitive consistency introduces path dependence
- The agent stops search earlier
- Posterior beliefs are stickier

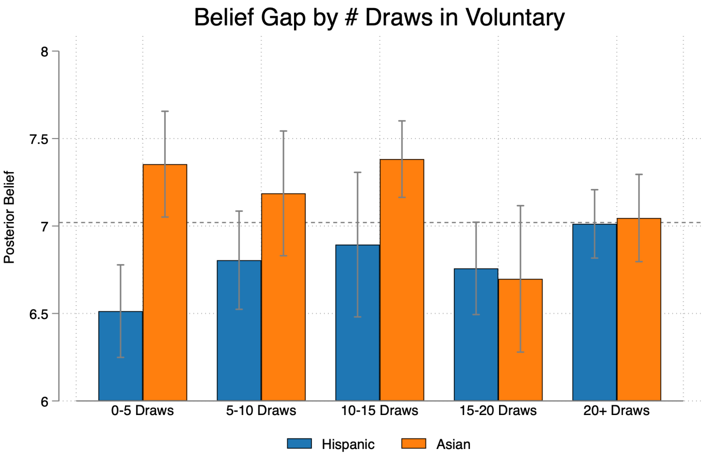
▸ Back to Theory

Ex-Ante Commitment Under Cognitive Consistency

$$\max_{\tau \leq T} \mathbb{E}_0 \left[u(\mu_\tau, \theta) + \sum_{t=1}^{\tau} \gamma v(\mu_0, \mu_t) - c\tau \right] \quad (7)$$

- No longer equivalent to the sequential decision problem
- Sample less than the sequential agent: $\mathbb{E}[\tau^{commit}] \leq \mathbb{E}[\tau^{sequential}]$
- Posteriors more biased towards priors than sequential agent

Belief Gap in Voluntary by Number of Draws



▸ Voluntary Posteriors